

Exit Exam Model Mock 4

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#networking

1. In a _____ relationship, one device controls traffic and the others must transmit through it?
 - A. Peer-to-peer
 - B. Point-to-point
 - C. Primary-secondary
 - D. Full duplex
2. Which one of the following is different from the other?
 - A. 172.16.35.12
 - B. 188.17.12.14
 - C. 130.17.12.24
 - D. 126.17.12.254
3. Which of the following is a networking technique that enables us to transfer two or more different data signals simultaneously on the same communication channel?
 - A. Switching
 - B. Layering task
 - C. Multiplexing
 - D. None
4. What is the prefix length notation for the subnet mask 255.255.255.248?
 - A. /30
 - B. /27
 - C. /28
 - D. /29
5. The wireless LAN specification is defined by IEEE, called, _____ which covers the data link and physical layer?
 - A. IEEE 802.2
 - B. IEEE 802.11
 - C. IEEE 802.3
 - D. IEEE 802.5
6. A network where there is no resource reservation is called ?
 - A. Circuit-switched networks
 - B. Message-switched networks
 - C. Packet-switched networks
 - D. Satellite network
7. The whole communication between client and server to transmit a file may take ?
 - A. 2RTT
 - B. RTT + TRANSMIT TIME
 - C. 2RTT + TRANSMIT TIME
 - D. RTT

8. Most networks employ devices for routing services. Routers work at which of the following OSI layers?
- A. Transport
 - B. Network
 - C. Presentation
 - D. Session

#database

9. Databases threats that are related to unauthorized disclosure are called?
- A. Loss of integrity
 - B. Loss of availability
 - C. Loss of confidentiality
 - D. None
10. Which of these statements about serializable schedules is true?
- A. Every serializable schedule contains no conflicting actions.
 - B. Every serializable schedule is recoverable.
 - C. Every 2pl schedule is serializable.
 - D. None of the above
11. Concurrency problem that occurs when one transaction is allowed to see the intermediate results of another transaction before it has committed, which will be failed for some reason is called?
- A. Incorrect summery problem
 - B. Lost update problem
 - C. Dirty read problem
 - D. All
 - E. None
12. The purpose of database recovery is ?
- A. To bring the database into the last consistent state.
 - B. To preserve consistency of transactions.
 - C. To preserve durability property of transactions.
 - D. All
 - E. None
13. Which of these statements about recoverable schedules is true?
- A. In a recoverable schedule, if a transaction t commits, then any other transaction that t read from must also have committed.
 - B. Every recoverable schedule is serializable.
 - C. In a recoverable schedule, no transaction will ever be aborted because a transaction that it read from has aborted.
 - D. None of the above.
14. What is a database?
- A. A collection of information organized in such a way that a computer program can quickly select desired pieces of data
 - B. A collection of unorganized data
 - C. A collection of documents
 - D. none of the above

15. Which language is used to query databases?

- A. C++
- B. python
- C. SQL
- D. Java

#internet programming

16. What two things do you need to create Web Pages & view them?

- A. A text editor and a web browser
- B. A text editor and a compiler
- C. A compiler and a web browser
- D. All of the above

17. Where in an HTML document is the correct place to refer to an external style sheet?

- A. In the <head> section
- B. At the top of the document
- C. In the <body> section
- D. At the end of the document

18. What CSS define in HTML?

- A. How to save html elements
- B. How to display html elements
- C. How to send html elements
- D. How to made html elements

19. What does an HTML tag do?

- A. It specifies formatting and layout instructions for your web page.
- B. It hides programming instructions from view.
- C. It determines the organizational structure of your web site.
- D. It connects your web site to an operating environment.

20. Which of the following is not the scope of Variable in PHP?

- A. Local
- B. Global
- C. Static
- D. Extern

21. A function in PHP which starts with __ (double underscore) is known as?

- A. Inbuilt function
- B. Default function
- C. Magic function
- D. User defined function

22. Where *setcookie()* function must appear in PHP?

- A. Before tag
- B. After tag
- C. In tag
- D. Anywhere

#Software engineering

26 Which of the following requirement represents the existence of the element over time?

- A. Lifeline
- B. Messages
- C. Communication
- D. Activation bar

27. Which of the following is derived from the analysis model scenario-based elements, flow-oriented elements, and behavioral elements?

- A. Component-level design
- B. Interface design
- C. Data/class design
- D. Architectural design

28. Which of the following is created by transforming the analysis model class-based elements into classes and data structures required to implement the software?

- A. Component-level design
- B. Interface design
- C. Data/class design
- D. Architectural design

29. Which Diagram emphasize on features that do not change with time ?

- A. Dynamic
- B. Structural
- C. Collaboration
- D. Activity

30. Which of the following Diagram focuses on the individual concepts that are manipulated by the system, their properties, and their relationships?
- A. Class based modelling
 - B. Structural
 - C. Collaboration
 - D. Activity
31. Which of the following is invalid relationships in Use Case Diagrams?
- A. Use
 - B. Generalization
 - C. Subtyping
 - D. Include
 - E. Extend
32. Which diagram in UML shows a complete or partial view of the structure of a modeled system at specific time ?
- A. sequence diagram
 - B. collaboration diagram
 - C. object diagram
 - D. state chart diagram
33. An activity diagram is used to illustrate one of the following?
- A. The flow of a complicated use case.
 - B. A workflow across use cases.
 - C. The logic of an algorithm.
 - D. All
 - E.

#project management

34. What is the research-oriented software development?
- A. Product-related problems
 - B. Process-related problems
 - C. People-related problems
 - D. Technology-related problems
 - E. None

35. Technical kernel of software engineering ?
- A. Testing
 - B. Requirement analysis
 - C. Design
 - D. Implementation
36. What is the preferred order for performing testing on information technology projects?
- A. Unit testing, integration testing, system testing, user acceptance testing
 - B. Unit testing, system testing, integration testing, user acceptance testing
 - C. Unit testing, system testing, user acceptance testing, integration testing
 - D. Unit testing, integration testing, user acceptance testing, system testing
37. Which of the following is not a process in project communications management?
- A. Planning communications management
 - B. Controlling communications
 - C. Managing communications
 - D. Managing stakeholders
38. _____ is an uncertainty that can have a negative or positive effect on meeting project objectives?
- A. Risk utility
 - B. Risk tolerance
 - C. Risk management
 - D. Risk
39. During project execution, a team member comes to the project manager, as he is not sure of what work he needs to accomplish on the project. Which of the following documents contain detailed descriptions of work packages?
- A. Scope management plan
 - B. Project scope statement
 - C. Wbs
 - D. Activity list

40. Which of the following is related to the overall functionality of the delivered software?
- A. Product-related metrics
 - B. Function-related metrics
 - C. Size-related metrics
 - D. None of the mentioned

#machine learning

41. Which machine learning algorithm is suitable for solving regression problems?
- A. K-means Clustering
 - B. Random Forest
 - C. K-Nearest Neighbors (KNN)
 - D. Apriori Algorithm
42. What does "overfitting" refer to in machine learning?
- A. When a model performs well on the training data but poorly on new data
 - B. When a model performs well on new data but poorly on the training data
 - C. When a model perfectly fits the training data
 - D. When a model is too simple to capture the underlying patterns
43. Which type of machine learning algorithm aims to mimic the process of human learning?
- A. Supervised learning
 - B. Unsupervised learning
 - C. Reinforcement learning
 - D. Deep learning
44. What is the purpose of the validation set in machine learning?
- A. To train the model
 - B. To fine-tune hyper parameter
 - C. To test the model's generalization
 - D. To provide additional training data
45. Which of the following is a supervised learning task?
- A. Clustering
 - B. Reinforcement learning
 - C. Dimensionality reduction
 - D. Classification
46. In machine learning, what is a model?
- A. A physical machine that performs computations
 - B. A representation of the data

- C. A set of training examples
- D. A type of computer algorithm

#AI

- 47. A computer program that contains expertise in a particular domain is called?
 - A. Intelligent planner
 - B. Automatic processor
 - C. Expert system
 - D. Operational symbolizer
- 48. An Artificial Intelligence technique that allows computers to understand associations and relationships between objects and events is called?
 - A. Heuristic processing
 - B. Cognitive science
 - C. Relative symbolism
 - D. Pattern matching
- 49. Which of the following is the component of the learning system?
 - A. Goal
 - B. Model
 - C. Learning rules
 - D. All of the mentioned
- 50. Classifying email as spam, labelling webpages based on their content, and voice recognition are examples of?
 - A. Supervised learning
 - B. Unsupervised learning
 - C. Machine learning
 - D. Deep learning
- 51. The field that investigates the mechanics of human intelligence is?
 - A. History
 - B. Cognitive science
 - C. Psychology
 - D. Sociology
- 52. What is the goal of artificial intelligence?
 - A. To solve real-world problems
 - B. To solve artificial problems
 - C. To explain various sorts of intelligence
 - D. To extract scientific causes

#operating system

53. Which of the following is NOT a function of an operating system?
- A. File management
 - B. memory management
 - C. Data analysis
 - D. process management
54. What is the purpose of the kernel in an operating system?
- A. To manage the computer's hardware
 - B. To provide a graphical user interface
 - C. To run application software
 - D. to store files and documents
55. Which of the following is NOT a type of operating system?
- A. Real-time
 - B. Batch processing
 - C. hierarchical
 - D. time-sharing
56. What is virtual memory in an operating system?
- A. A memory management technique
 - B. A type of physical memory
 - C. A form of cache memory
 - D. A hardware component
57. Which of the following is an example of a multi-user operating system?
- A. Windows XP
 - B. Windows 10 home edition
 - C. Linux
 - D. Mac OS
58. What is the purpose of device drivers in an operating system?
- A. To manage hardware devices
 - B. To edit documents
 - C. To run application software
 - D. To connect to the internet
59. In the short term and the long term scheduler the primary distinction between is?
- A. The frequency of their execution
 - B. The length of their queues
 - C. The type of processes they schedule
 - D. None
60. The two steps the operating system takes to use a disk to hold its files are ____ and ____?
- A. Caching & logical formatting
 - B. Logical formatting & swap space creation

- C. Swap space creation & caching
- D. Partitioning & logical formatting

#mobile application development

61. Which of the following components is responsible for managing the user interface of an Android app?
- A. Activity
 - B. Intent
 - C. Service
 - D. Broadcast receiver
62. Which file defines the user interface layout of an Android app?
- A. xml file
 - B. java file
 - C. gradle file
 - D. json file
63. Which method is called when an Activity is first created in Android?
- A. onCreate()
 - B. onStart()
 - C. onResume()
 - D. onInit()
64. What is the purpose of the AndroidManifest.xml file?
- A. It defines the layout of the app's user interface.
 - B. It contains configuration information about the app.
 - C. It stores the source code of the app.
 - D. It handles the database operations of the app.
65. Which of the following components is used to perform long-running operations in the background without interacting with the UI?
- A. Activity
 - B. Service
 - C. Intent
 - D. Broadcast receiver
66. What is the purpose of the 'adb' tool in Android development?
- A. To compile the android app code.
 - B. To debug the android app.
 - C. To manage the android device's storage.
 - D. To design the user interface of the android app.

67. Which of these views concerned with the functionality that the system provides to end-users?
- A. Logical view
 - B. Process view
 - C. Development view
 - D. Physical view
68. Functionality does determine architecture?
- A. True
 - B. False
69. The following can be modifiability tactics except?
- A. Localize changes
 - B. Prevention of ripple effect
 - C. Defer binding time
 - D. Resource arbitration
70. There was a time when every household, town, farm or village had its own water well. Today, shared public utilities give us access to clean water by simply turning on the tap_____ works in a similar fashion?
- A. Cloud computing
 - B. Hdfs
 - C. Map reduce
 - D. Broker
 - E. MVC
71. a design pattern which provides a unified interface to a set of interfaces in a subsystem without damaging the generic form of the subsystem?
- A. Adapter
 - B. Façade
 - C. Factory
 - D. Iterator
72. Which one of the following statements is not true about software Architecture?
- A. It defines software elements.
 - B. Systems can and do comprise more than one structure, all are part of the architecture.
 - C. Every computing system with software has an architecture.
 - D. The behavior of each element is part of the architecture as far as its behavior can be observed or discerned from the point of view of another element.
 - E. None
73. Which one of the following measures the relative functional strength of a module?
- A. Coupling
 - B. Cohesion
 - C. Pattern
 - D. Software procedure

#object oriented programming

74. In Object-Oriented Programming, what is the term used to describe the ability of one class to inherit properties and behaviors from another class?
- A. Abstraction
 - B. Polymorphism
 - C. Encapsulation
 - D. Inheritance
75. Which is not true about the object-oriented approach?
- A. Emphasis is on data rather than procedure
 - B. Data is hidden and cannot be accessed by external functions
 - C. Objects communicate through method
 - D. It supports abstract data but not the class
76. Which of the following feature is also known as run-time binding or late binding?
- A. Static binding
 - B. Dynamic loading
 - C. Dynamic binding
 - D. Data hiding
77. Which among the following is not a necessary condition for constructors?
- A. Its name must be same as that of class
 - B. It must not have any return type
 - C. It must contain a definition body
 - D. It can contains arguments
78. What is encapsulation in OOP?
- A. It is a way of combining various data members and member functions that operate on those data members into a single unit
 - B. It is a way of combining various data members and member functions into a single unit which can operate on any data
 - C. It is a way of combining various data members into a single unit
 - D. It is a way of combining various member functions into a single unit
79. Which of the following best describes member function overriding?
- A. Member functions having the same name in derived class only
 - B. Member functions having the same name and different signature inside main function
 - C. Member functions having the same name and signature in base and derived classes
 - D. Member functions having the same name in base class only

#fundamental of programming

80. Computer programming is the process of performing a particular computation, usually by designing/ building an executable computer is?
- A. Code
 - B. Program
 - C. File
 - D. All of the above
81. The source of a program is written in one or more languages is directly executed by?
- A. Arithmetic and logic unit
 - B. Central processing unit
 - C. Main storage
 - D. None of the above
82. Which keyword is used by method to refer to the object that invoked it?
- A. This
 - B. Catch
 - C. Abstract
 - D. Import
83. Which of the following method of class String is used to obtain length of String object in Java?
- A. Length()
 - B. Sizeof()
 - C. Lengthof()
 - D. Limitof()
84. Which is more effective while calling the C++ functions?
- A. Call by object
 - B. Call by pointer
 - C. Call by value
 - D. Call by reference
85. What does the "sizeof" operator return in C++ when used with an array?
- A. The number of elements in the array.
 - B. The size of each element in the array.
 - C. The total size of the array in bytes.
 - D. The memory address of the first element in the array.

86. Which is more effective while calling the C++ functions?
- A. Call by object
 - B. Call by pointer
 - C. Call by value
 - D. Call by reference
87. Which of the following is not a type of Constructor in C++?
- A. Default constructor
 - B. Parameterized constructor
 - C. Copy constructor
 - D. Friend constructor
88. What is the purpose of a structure in C++?
- A. To store a collection of elements of the same data type.
 - B. To store a collection of elements of different data types.
 - C. To perform arithmetic calculations.
 - D. To define a loop in a program.
89. Which of the following statements is true about structures in C++?
- A. Structures cannot have functions as members.
 - B. Structures are always passed by value when used as function arguments.
 - C. Structures cannot contain arrays as members.
 - D. Structures automatically initialize their members to zero.
90. What does the "->" operator do in C++ when used with a structure pointer?
- A. It performs bitwise and operation.
 - B. It performs logical or operation.
 - C. It accesses a member of a structure through a pointer.
 - D. It dereferences a pointer to a structure.

#Data structure

91. Which Algorithm cannot use Divide and Conquer Principle?
- A. Merge sort
 - B. Binary search
 - C. Bubble sort
 - D. All
92. Which data structure is suitable to represent hierarchical relationship between elements?
- A. Graph
 - B. Tree
 - C. Dequeue
 - D. Priority

93. Which traversal of a binary search tree display elements in ascending order?
- A. Inorder traversal
 - B. Preorder traversal
 - C. Postorder traversal
 - D. All
94. Which data structure is more appropriate to implement recursive algorithms?
- A. Tree
 - B. Linked list
 - C. Queue
 - D. Stack
95. Which one of the following is Operation of a stack?
- A. Insert
 - B. Push
 - C. Add
 - D. Enqueue
 - E. None
96. The data structure required to evaluate a postfix expression is ?
- A. Queue
 - B. Stack
 - C. Linked-list
 - D. Array
97. A linear list in which the last node points to the first node is?
- A. Singly linked list
 - B. Doubly linked list
 - C. Circular linked list
 - D. Tree

#software security

98. Cryptography intercepts ____ type of data accessibility?
- A. Authorized
 - B. Unauthorized
 - C. Legitimate
 - D. All
99. How many keys do Triple DES operates with?
- A. 2
 - B. 4
 - C. 3
 - D. 5

100. Which folder contains Junk emails?
- A. Inbox
 - B. Unwanted
 - C. Spam
 - D. None
101. Which of the following algorithm has overcome triple DES?
- A. AES
 - B. BLOW FISH
 - C. DSA
 - D. All
102. SMTP stands for?
- A. Simple Mail Text Processing
 - B. Secure Mail Transfer Protocol
 - C. Simple Mail Transfer Protocol
 - D. Secure Message Transfer Process
103. Which of the following data security principle does checksum verifies?
- A. INTEGRITY
 - B. AUTHENTICITY
 - C. CONFIDENTIALITY
 - D. REPUDIABILITY

#software testing

104. Which of the following is not a primary goal of software testing?
- A. Finding defects in the software
 - B. Ensuring the software meets user requirements
 - C. Improving the performance of the software
 - D. Verifying that the software functions correctly
105. Which testing technique involves executing the software with known inputs and comparing the actual results with expected results?
- A. White-box testing
 - B. Black-box testing
 - C. Unit testing
 - D. Regression testing
106. What is the purpose of regression testing?
- A. To test a new feature or functionality.
 - B. To ensure that previously developed and tested software still performs correctly after changes or updates.
 - C. To test the software from end to end.
 - D. To test the performance of the software under load.

107. Which testing technique examines the internal structure or workings of a software application?
- A. White-box testing
 - B. Black-box testing
 - C. Integration testing
 - D. User acceptance testing
108. What is the purpose of smoke testing?
- A. To test the basic functionality of the software
 - B. To test the software's ability to handle a high volume of users
 - C. To test the software's performance under stress
 - D. To test the software's compatibility with different operating systems
109. Which testing technique involves testing individual units or components of a software application in isolation?
- A. System testing
 - B. Integration testing
 - C. Unit testing
 - D. Acceptance testing
110. What is the purpose of usability testing?
- A. To test the software's ability to handle a high volume of users.
 - B. To test the software's performance under stress.
 - C. To evaluate how easy and intuitive the software is to use.
 - D. To test the software's compatibility with different operating systems.
111. What does the "bug severity" refer to in software testing?
- A. The likelihood of a bug occurring
 - B. The impact of a bug on the software's functionality
 - C. The number of bugs present in the software
 - D. The time required to fix a bug

_____ The End _____